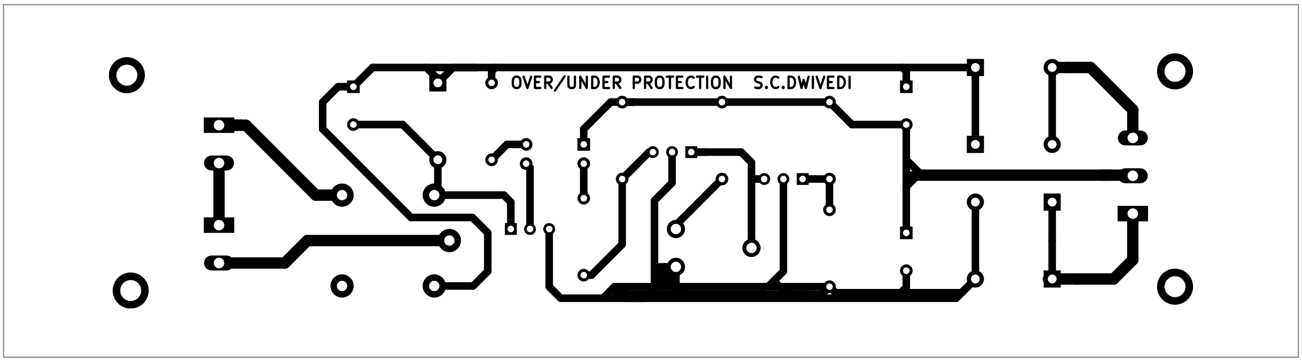

Fig. 3: Actual-size PCB layout

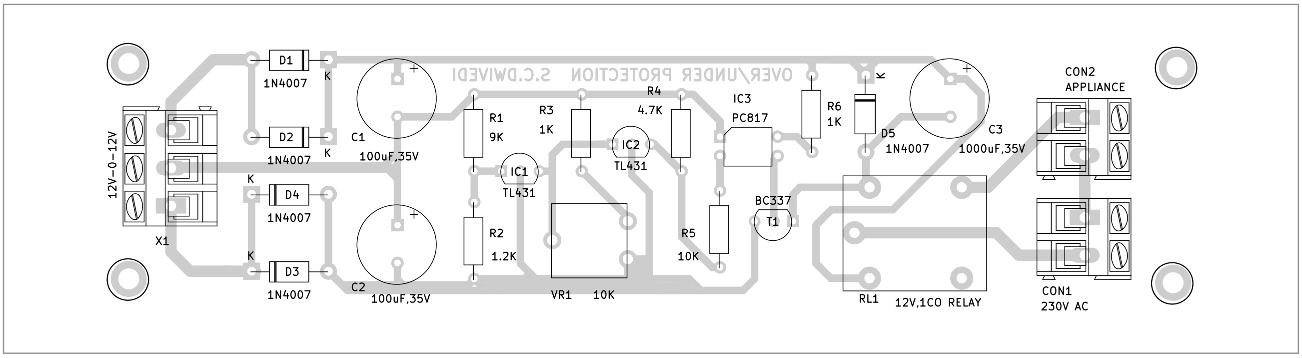

Fig. 4: Components side of PCB

the relay. Resistor R6 biases the internal transistor of the optocoupler and resistor R4 is used to bias the internal transistor of IC2.

Secondary voltage of transformer X1 proportionally changes as the primary (AC mains) voltage changes. This change reflects at the external reference pins (pin 1) of IC1 and IC2. The R1-R2 resistor network sets the pin 1 voltage below the internal reference voltage of IC1.

IC1 acts as an open circuit that exists between cathode pin 3 and anode pin 2 of optocoupler IC3. Preset VR1 is adjusted to set the pin 1 voltage of IC2 above the internal voltage reference of IC2. In this case, IC2 acts as a short circuit and so the optocoupler keeps conducting.

The optocoupler output biased through R6 is fed into the base of transistor T1, which drives relay RL1 to on state. If the voltage increases and becomes greater than the internal reference voltage at pin 1 of IC1, the optocoupler starts conducting between pin 3 and pin 2 and acts as a short circuit. This sets IC2's reference pin 1 below the internal reference voltage and forces IC2 to an open-circuit state, and so the optocoupler and relay turn off and switch off the load. The same happens when the voltage decreases at pin 1 of IC2 due to a fall in AC mains voltage.

Construction and calibration

An actual-size, single-side PCB layout for the circuit is shown in Fig. 3 and its component layout is shown in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix VR1 outside the box after proper calibration (if using a pot in place of preset). Fit the transformer inside the box.

For the calibration, you may use a variac (variable transformer) for setting the over voltage and under voltage limits. The over voltage limit has been set around 270V with the help of resistors R1 and R2, which you can change (if required) by changing values of these two resistors.

For the under voltage limit, set the variac to around 180V and set the preset in such a way that the relay energises at this voltage. This completes the calibration.

After calibration, enclose the box and connect the 230V AC mains voltage to the primary of transformer X1. Connect the AC mains and the appliance to be protected across connectors CON1 and CON2, respectively. Now your circuit is ready to use. EFY

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